

Create hands-on practice questions for data visualization aesthetics

Absolutely — here are hands-on practice questions for Data Visualization Aesthetics, designed to help you practice choosing and applying visual properties such as color, size, shape, position, labels, and transparency.

Hands-on Practice Questions

Color

Create a bar chart showing sales for 5 products.

Use a different color for each product.

Then redesign it using a single color with different intensities.

Which version makes comparison easier, and why?

Size

Create a scatter plot showing advertising spend vs. sales for 10 companies.

Use point size to represent the number of customers.

Explain what additional information the size aesthetic provides.

Shape

Create a scatter plot of student study hours vs. exam scores.

Use different shapes for students from different departments.

Which department appears to have the strongest relationship?

Position

Create a bar chart comparing the population of 8 cities.

Arrange the bars from smallest to largest.

Create another version in alphabetical order.

Which arrangement makes comparison easier?

Color + Position

Visualize monthly sales for three products using a line chart.

Use color to distinguish products.

Add data labels only to the highest and lowest sales points.

Identify which aesthetic makes the trend easiest to follow.

Transparency

Create a scatter plot containing at least 100 observations.

First plot the points with full opacity.

Then reduce their transparency.

Explain how transparency helps reveal overlapping observations.

Avoiding Misleading Aesthetics

Create a bar chart comparing sales of five stores.

Make one version with a y-axis starting at 0 and another starting at 80.

Compare the visual impression.

Which version is more appropriate? Explain why.

Choosing the Right Aesthetic

For each variable below, decide which aesthetic—position, color, size, shape, or transparency—would be most appropriate:

Product category

Revenue

Number of customers

Region

Customer satisfaction score

Year

Improve a Poor Visualization

Design a chart containing at least three unnecessary visual elements, such as excessive colors, 3D effects, heavy borders, or too many labels.

Redesign it using better aesthetics.

List the changes you made and explain why.

Mini Project

Using a dataset of your choice:

Create a visualization using at least 4 aesthetics.

Use position for one variable.

Use color for another.

Use size or shape for a third.

Add appropriate titles, axis labels, and a legend.

Write 3 observations from your final visualization.

Challenge Question ★

Create two different visualizations of the same dataset. Use different aesthetics in each one and answer:

Which visualization communicates the main message more effectively, and what role do the visual aesthetics play in making it easier to understand?

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If you're practicing R/ggplot2, Python/Seaborn/Matplotlib, or Tableau, I can also turn these into lab-style questions with datasets, expected output, and solutions.

Yes

Great. Here's a hands-on lab sheet for Data Visualization Aesthetics using Python (Matplotlib + Seaborn). It progresses from basic to challenging.

Data Visualization Aesthetics — Practical Lab

Dataset

Use this dataset for Questions 1–10:

```
import pandas as pd
```

```
data = {  
    "Product": ["Laptop", "Phone", "Tablet", "Monitor", "Keyboard", "Mouse"],  
    "Sales": [120, 250, 180, 90, 300, 220],  
    "Profit": [30, 55, 40, 20, 75, 45],  
    "Customers": [80, 150, 110, 60, 200, 170],  
    "Rating": [4.5, 4.2, 4.4, 4.0, 4.7, 4.3],  
}
```

```
"Category": ["Electronics", "Electronics", "Electronics",  
            "Electronics", "Accessories", "Accessories"]  
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

Q1. Position Aesthetic — Basic

Create a bar chart showing Sales for each Product.

Tasks:

Product → x-axis

Sales → y-axis

Add a title and axis labels.

Identify the product with the highest sales.

Concept: Position is one of the most important visual aesthetics because it allows accurate comparison.

Q2. Color Aesthetic

Create a bar chart of Sales.

Tasks:

Give each product a different color.

Add a meaningful title.

Explain whether using multiple colors improves the visualization.

Concept: color can be used to distinguish categories.

Q3. Size Aesthetic

Create a scatter plot using:

Sales → x-axis

Profit → y-axis

Customers → point size

Tasks:

Add appropriate axis labels.

Add a title.

Identify which product has the largest point.

Concept: Size can represent a third quantitative variable.

Q4. Shape Aesthetic

Create a scatter plot of:

Sales → x-axis

Profit → y-axis

Category → marker/shape

Tasks:

Use different markers for Electronics and Accessories.

Add a legend.

Describe any difference between the two categories.

Challenge: Try creating the same plot using Seaborn.

Q5. Color + Size

Create a scatter plot where:

x = Sales

y = Profit

color = Category

size = Customers

Question:

What additional information can you understand from this visualization compared with a simple scatter plot?

Q6. Transparency

Create a scatter plot with:

Sales vs Profit

Use different transparency (alpha) values such as:

alpha=1.0

alpha=0.5

alpha=0.2

Tasks:

Compare the three plots.

Explain when transparency is useful.

Explain why transparency is particularly useful when many points overlap.

Q7. Labels and Annotations

Create a scatter plot of Sales vs Profit.

Add the product name next to every point.

Challenge:

Highlight the product with the highest profit using an annotation.

Q8. Avoid Misleading Visualization ⚠️

Create two bar charts showing product sales.

Chart A:

y-axis starts at 0.

Chart B:

y-axis starts at 200.

Compare the two charts.

Questions:

Which chart gives a more accurate visual impression?

How can changing the axis range mislead the viewer?

When might a non-zero baseline be acceptable?

Q9. Aesthetic Makeover

Create a deliberately poor visualization containing:

Too many colors

Large unnecessary title

Rotated labels

Excessive gridlines

Unnecessary legend

Then redesign it.

Submit:

Before visualization

After visualization

Five changes you made

Explanation of why the second version is better

Q10. Final Mini Project ★

Create a visualization using at least four aesthetics.

Your visualization must use:

Variable	Aesthetic
----------	-----------

Sales	Position
-------	----------

Profit	Position
--------	----------

Category	Color
----------	-------

Customers	Size
-----------	------

Product	Label
---------	-------

Requirements

Your final chart should have:

Clear title

Proper axis labels

Legend where appropriate

Appropriate colors

Appropriate scale

Readable labels

No unnecessary decoration

Analysis

Write 5 observations about your visualization.